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**TP - 02
Homework**

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Lab02.1: Hello One's Name

Implement an application Java that displays “Hello <input_name>!” where <input_name> represents user input from keyboard.

```
1  package Lecture_2;
2
3  import java.util.Scanner;
4
5  public class Exercise1 {
6
7      Run | Debug
      public static void main(String[] args) {
8          Scanner scan = new Scanner(System.in);
9          System.out.print(s:"Input your name: ");
10         String name = scan.nextLine();
11         System.out.println("Hello " + name + "!");
12         scan.close();
13     }
14 }
15
16 }
```

```
Input your name: Nyheng
Hello Nyheng!
PS C:\Users\User\Documents\I2\OOP\TP\src>
```

Lab02.2: Currency Converter

Write a program in Java to money in Riels to Dollar. Suppose that conversion rate is 1\$ = 4000€.

```
Program for converting money in Riels to Dollars.
Please input a conversion rate of 1 dollar in Riel: 4000
Please input money to exchange in Riels: 2500
2500.0 RIELS = 0.625 USD
PS C:\Users\User\Documents\I2\OOP\TP\src>
```



```

==== Covid-19 Screen Health ====
Q1. Feeling feverish and/or having chills?(Yes/No): Yes
Q2. Has there been any use of fever reducing medication within the last 24 hours not due to another health condition?(Yes/No): No
Q3. A new cough that is not due to another health condition?(Yes/No): Yes
Q4. New chills that are not due to another health condition?(Yes/No): No
Q5. A new sore throat that is not due to another health condition?(Yes/No): Yes
Q6. A new loss of taste or smell?(Yes/No): No
Q7. Have you had a positive test for the virus that causes COVID-19 disease within the past 10 days?(Yes/No): Yes
Q8. In the past 14 days, have you had close contact (within about 6 feet for 15 minutes or more) with someone with suspected or confirmed COVID-19?(Yes/No): No
=====
| Question          | Answer          |
=====
| Q1                | Yes             |
-----
| Q2                | No              |
-----
| Q3                | Yes             |
-----
| Q4                | No              |
-----
| Q5                | Yes             |
-----
| Q6                | No              |
-----
| Q7                | Yes             |
-----
| Q8                | No              |
-----
PS C:\Users\User\Documents\I2\OOP\TP\src>

```

Lab02.4: Seconds to Times

Implement an application Java that take a given number of seconds for converting into time with format.

Hours: Minutes: Seconds.

```

1  package Lecture_2;
2
3  import java.util.Scanner;
4
5  public class Exercise4 {
6      public static void Conversion(int number) {
7          int hours = number / 3600;
8          int remainderMinutes = number % 3600;
9          int minutes = remainderMinutes / 60;
10         int seconds = remainderMinutes % 60;
11         System.out
12         .println("Time corresponding to " + number + " seconds is:: " + hours + ":" + minutes + ":" + seconds);
13     }
14
15     Run | Debug
16     public static void main(String[] args) {
17         Scanner scan = new Scanner(System.in);
18         System.out.print(s:"Input a number in second: ");
19         int number = scan.nextInt();
20         Conversion(number);
21         scan.close();
22     }
23 }

```

PROBLEMS 14 OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER 11

```

PS C:\Users\User\Documents\I2\OOP\TP\src> & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.6.7-hotspot\bin\java.exe' '-XX:+ShowCodeDetails' 'C:\Users\User\AppData\Roaming\Code\User\workspaceStorage\b9c1a0a7c14e649f4ec5862176279586\redhat.java\jdt_ws\src_6d53b3a\bin' '-m' '00'
Input a number in second: 3850
Time corresponding to 3850 seconds is:: 1:4:10

```

Lab02.5: Equation

Write a program in Java to calculate value of x in a formula below:

Where y and z are given by user from keyboard. Suppose that x, y, z are non-zero.

```
1  package Lecture_2;
2
3  import java.util.Scanner;
4
5  public class Exercise5 {
6      public static double FindX(double y, double z) {
7          double x = (y * z) / (y + z);
8          return x;
9      }
10
11     Run | Debug
12     public static void main(String[] args) {
13         Scanner scan = new Scanner(System.in);
14         double y = 0;
15         double z = 0;
16         System.out.println(x:"Program for calculating equation 1/x = 1/y + 1/z to find value of x");
17         while (true) {
18             System.out.print(s:"Please input y: ");
19             y = scan.nextDouble();
20             if (y != 0) {
21                 break;
22             }
23             System.out.println(x:"y should be non zero");
24         }
25         while (true) {
26             System.out.print(s:"Please input z: ");
27             z = scan.nextDouble();
28             if (z != 0) {
29                 break;
30             }
31             System.out.println(x:"Z should be non zero");
32         }
33         System.out.println("Result x: = " + FindX(y, z));
34     }
35 }
```

```
Program for calculating equation 1/x = 1/y + 1/z to find value of x
Please input y: 20
Please input z: 30
Result x: = 12.0
PS C:\Users\User\Documents\I2\OOP\TP\src>
```

Lab02.6: Prime Number

Prime number is positive number greater than 2 and divisible only to its own and 1.

Implement an application Java that let user input a number then determine if it is a prime number

```

1 package Lecture_2;
2
3 import java.util.Scanner;
4
5 public class Exercise6 {
6     public static void PrimeCheck(int number) {
7         if (number <= 1) {
8             System.out.println(number + " is not Prime number");
9         }
10        boolean isPrime = true;
11        for (int i = 2; i <= Math.sqrt(number); i++) {
12            if (number % i == 0) {
13                System.out.println(number + " is not a Prime Number, because it is divisible by " + i);
14                isPrime = false;
15                break;
16            }
17        }
18        if (isPrime) {
19            System.out.println(number + " is a Prime Number");
20        }
21    }
22
23    Run | Debug
24    public static void main(String[] args) {
25        Scanner scan = new Scanner(System.in);
26        System.out.print(s;"Input a number to check whether it is a Prime number: ");
27        int number = scan.nextInt();
28        PrimeCheck(number);
29        scan.close();
30    }
31 }

```

```

Input a number to check whether it is a Prime number: 5
5 is a Prime Number
PS C:\Users\User\Documents\I2\OOP\TP\src>

```

Lab02.7: Character Count

Write a program using to count number of Vowel, Consonant and Space:

```

Input a sentence: Hello Classmate
count of vowel is: 5
count of consonant is: 9
count of space is: 1
PS C:\Users\User\Documents\I2\OOP\TP\src>

```

```

1  package Lecture_2;
2
3  import java.util.Scanner;
4
5  public class Exercise7 {
6      public static void Analyse(String sentence) {
7          int spacecount = 0;
8          int vowelcount = 0;
9          int consonantcount = 0;
10         int size = sentence.length();
11         for (int i = 0; i < size; i++) {
12             if (sentence.charAt(i) == ' ') {
13                 spacecount++;
14             } else if (sentence.charAt(i) == 'a' || sentence.charAt(i) == 'e' || sentence.charAt(i) == 'i' ||
15                 sentence.charAt(i) == 'o' || sentence.charAt(i) == 'u' || sentence.charAt(i) == 'A' ||
16                 sentence.charAt(i) == 'E' || sentence.charAt(i) == 'I' || sentence.charAt(i) == 'O' ||
17                 sentence.charAt(i) == 'U') {
18                 vowelcount++;
19             } else {
20                 consonantcount++;
21             }
22         }
23         System.out.println("count of vowel is: " + vowelcount);
24         System.out.println("count of consonant is: " + consonantcount);
25         System.out.println("count of space is: " + spacecount);
26     }
27
28     Run | Debug
29     public static void main(String[] args) {
30         Scanner scan = new Scanner(System.in);
31         System.out.print(s:"Input a sentence: ");
32         String sentence = scan.nextLine();
33         Analyse(sentence);
34         scan.close();
35     }

```

Lab02.8: Minimum Value

Write a program to find a minimum value among A, B, C, D and E

```

Input the value of A: 5
Input the value of B: 10
Input the value of C: -20
Input the value of D: 30
Input the value of E: 1
The smallest number among A, B, C, D and E is: -20.0
PS C:\Users\User\Documents\I2\OOP\TP\src>

```

```

1  package Lecture_2;
2
3  import java.util.Scanner;
4
5  public class Exercise8 {
6      public static void FindMinimum(double a, double b, double c, double d, double e) {
7          double min = a;
8          if (b < min)
9              min = b;
10         if (c < min)
11             min = c;
12         if (d < min)
13             min = d;
14         if (e < min)
15             min = e;
16         System.out.println("The smallest number among A, B, C, D and E is: " + min);
17     }
18
19     Run | Debug
20     public static void main(String[] args) {
21         Scanner input = new Scanner(System.in);
22         System.out.print(s:"Input the value of A: ");
23         double a = input.nextDouble();
24         System.out.print(s:"Input the value of B: ");
25         double b = input.nextDouble();
26         System.out.print(s:"Input the value of C: ");
27         double c = input.nextDouble();
28         System.out.print(s:"Input the value of D: ");
29         double d = input.nextDouble();
30         System.out.print(s:"Input the value of E: ");
31         double e = input.nextDouble();
32         FindMinimum(a, b, c, d, e);
33         input.close();
34     }

```

Lab2.9: Maximum value

Write a program to find a maximum value among all values in an Array. Users can input values in Array as many as they want.

```

Input the value in Array at Index 0: 5
Do you want to add more?(Yes/No): Yes
Input the value in Array at Index 1: 3
Do you want to add more?(Yes/No): Yes
Input the value in Array at Index 2: 20
Do you want to add more?(Yes/No): No
The biggest number among in array is: 20
PS C:\Users\User\Documents\I2\OOP\TP\src>

```



```

1  package Lecture_2;
2
3  import java.util.ArrayList;
4  import java.util.Scanner;
5
6  public class Exercise9 {
7      public static void FindMax(ArrayList<Integer> numbers) {
8          if (numbers.isEmpty()) {
9              System.out.println(x:"No number in array list");
10             return;
11         }
12         int max = numbers.get(index:0);
13         for (int i : numbers) {
14             if (i > max) {
15                 max = i;
16             }
17         }
18         System.out.println("The biggest number among in array is: " + max);
19     }
20
21     Run | Debug
22     public static void main(String[] args) {
23         Scanner scan = new Scanner(System.in);
24         ArrayList<Integer> numbers = new ArrayList<>();
25         int index = 0;
26         while (true) {
27             System.out.print("Input the value in Array at Index " + index + ": ");
28             int value = scan.nextInt();
29             numbers.add(value);
30             scan.nextLine();
31             System.out.print(s:"Do you want to add more?(Yes/No): ");
32             String choice = scan.nextLine();
33             if (choice.equals(anObject:"Yes")) {
34                 index++;
35             } else if (choice.equals(anObject:"No")) {
36                 break;
37             } else {
38                 System.out.println(x:"Incorrect Input!!");
39             }
40         }
41         FindMax(numbers);
42     }

```

Lab2.10: Math Menu

Write a program to choose menu and perform calculation based on options provided; if user choose an incorrection option, program still keep asking to re-input a correct option number.


```

45     input.nextLine();
46     System.out.println();
47     System.out.println("Answer of A + B = " + Addition(a, b));
48     System.out.println();
49     System.out.println(x:"Do you want to Continue? If no go to Menu.");
50     System.out.print(s:"Input your answer (y/n): ");
51     String ans = input.nextLine();
52     if (ans.equals(anObject:"y"))
53     |     continue;
54     else if (ans.equals(anObject:"n"))
55     |     break;
56     else {
57     |     System.out.println(x:"Incorrect Input!");
58     |     }
59     }
60     break;
61 case 2:
62     while (true) {
63     |     System.out.println(x:"=== Subtraction (-)===");
64     |     System.out.print(s:"Please Input the value A: ");
65     |     double a = input.nextDouble();
66     |     System.out.print(s:"Please Input the value B: ");
67     |     double b = input.nextDouble();
68     |     input.nextLine();
69     |     System.out.println();
70     |     System.out.println("Answer of A - B = " + Subtraction(a, b));
71     |     System.out.println();
72     |     System.out.println(x:"Do you want to Continue? If no go to Menu.");
73     |     System.out.print(s:"Input your answer (y/n): ");
74     |     String ans = input.nextLine();
75     |     if (ans.equals(anObject:"y"))
76     |     |     continue;
77     |     else if (ans.equals(anObject:"n"))
78     |     |     break;
79     |     else {
80     |     |     System.out.println(x:"Incorrect Input!");
81     |     |     }
82     |     }
83     break;
84 case 3:
85     while (true) {
86     |     System.out.println(x:"=== Multiplication (x)===");
87     |     System.out.print(s:"Please Input the value A: ");

```

```

88     double a = input.nextDouble();
89     System.out.print(s:"Please Input the value B: ");
90     double b = input.nextDouble();
91     input.nextLine();
92     System.out.println();
93     System.out.println("Answer of A x B = " + Multiplication(a, b));
94     System.out.println();
95     System.out.println(x:"Do you want to Continue? If no go to Menu.");
96     System.out.print(s:"Input your answer (y/n): ");
97     String ans = input.nextLine();
98     if (ans.equals(anObject:"y"))
99         continue;
100    else if (ans.equals(anObject:"n"))
101        break;
102    else {
103        System.out.println(x:"Incorrect Input!");
104    }
105    }
106    break;
107    case 4:
108        while (true) {
109            System.out.println(x:"=== Division (/)===");
110            System.out.print(s:"Please Input the value A: ");
111            double a = input.nextDouble();
112            System.out.print(s:"Please Input the value B: ");
113            double b = input.nextDouble();
114            input.nextLine();
115            System.out.println();
116            System.out.println("Answer of A / B = " + Division(a, b));
117            System.out.println();
118            System.out.println(x:"Do you want to Continue? If no go to Menu.");
119            System.out.print(s:"Input your answer (y/n): ");
120            String ans = input.nextLine();
121            if (ans.equals(anObject:"y"))
122                continue;
123            else if (ans.equals(anObject:"n"))
124                break;
125            else {
126                System.out.println(x:"Incorrect Input!");
127            }
128        }
129        break;

```

```

130        case 5:
131            System.out.println(x:"Thank you for using our program");
132            System.exit(status:0);
133            break;
134        default:
135            System.out.println(x:"Please try again!!");
136    }
137    }
138    }
139    }

```

```

=== Math Menu ===
1. Addition (+)
2. Subtraction (-)
3. Multiplication (x)
4. Division (/)
5. Quit
Choose an option: 1
=== Addition (+)===
Please Input the value A: 20
Please Input the value B: 30

Answer of A + B = 50.0

Do you want to Continue? If no go to Menu.
Input your answer (y/n): y
=== Addition (+)===
Please Input the value A: 30
Please Input the value B: 40

Answer of A + B = 70.0

Do you want to Continue? If no go to Menu.
Input your answer (y/n): n
=== Math Menu ===
1. Addition (+)
2. Subtraction (-)
3. Multiplication (x)
4. Division (/)
5. Quit
Choose an option: 2
=== Subtraction (-)===
Please Input the value A: 40
Please Input the value B: 30

Answer of A - B = 10.0

Do you want to Continue? If no go to Menu.
Input your answer (y/n): n

```

```

=== Math Menu ===
1. Addition (+)
2. Subtraction (-)
3. Multiplication (x)
4. Division (/)
5. Quit
Choose an option: 3
=== Multiplication (x)===
Please Input the value A: 2
Please Input the value B: 3

Answer of A x B = 6.0

Do you want to Continue? If no go to Menu.
Input your answer (y/n): n
=== Math Menu ===
1. Addition (+)
2. Subtraction (-)
3. Multiplication (x)
4. Division (/)
5. Quit
Choose an option: 4
=== Division (/)===
Please Input the value A: 4
Please Input the value B: 2

Answer of A / B = 2.0

Do you want to Continue? If no go to Menu.
Input your answer (y/n): n
=== Math Menu ===
1. Addition (+)
2. Subtraction (-)
3. Multiplication (x)
4. Division (/)
5. Quit
Choose an option: 5
Thank you for using our program
PS C:\Users\User\Documents\I2\OOP\TP\src>

```

Lab2.11: Class and Object

Create a model following objects:

- ID Card
- Facebook Post
- Phone
- Book

1. Implement a **Java Class** with **at least 5 attributes** defined
2. Create **at least 3 objects** with values assigned
3. Create **an array** and put all objects to that array

```

1  package Lecture_2;
2
3  class IDCard {
4      String id;
5      String name;
6      int age;
7      String gender;
8      String status;
9  }
10
11 class FaceBookPost {
12     int like;
13     int comment;
14     int share;
15     String type;
16     String genre;
17 }
18
19 class Phone {
20     String brand;
21     String model;
22     double price;
23     String color;
24     double weight;
25 }
26
27 class Book {
28     String isbn;
29     String title;
30     String author;
31     String genre;
32     double price;
33 }
34

```

```

35 public class Exercise11 {
    Run | Debug
36     public static void main(String[] args) {
37         IDCard idCard1 = new IDCard();
38         idCard1.id = "p20230028";
39         idCard1.name = "Chin Hongnyheng";
40         idCard1.age = 19;
41         idCard1.gender = "Male";
42         idCard1.status = "Student";
43
44         IDCard idCard2 = new IDCard();
45         idCard2.id = "p20230027";
46         idCard2.name = "John smith";
47         idCard2.age = 39;
48         idCard2.gender = "Male";
49         idCard2.status = "Teacher";
50
51         IDCard idCard3 = new IDCard();
52         idCard3.id = "p20230038";
53         idCard3.name = "Mary james";
54         idCard3.age = 49;
55         idCard3.gender = "Female";
56         idCard3.status = "Director";
57
58         IDCard idc[] = new IDCard[3];
59         idc[0] = idCard1;
60         idc[1] = idCard2;
61         idc[2] = idCard3;
62
63         FaceBookPost fb1 = new FaceBookPost();
64         fb1.like = 123;
65         fb1.comment = 124;
66         fb1.share = 29;
67         fb1.type = "photo";
68         fb1.genre = "Memes";
69
70         FaceBookPost fb2 = new FaceBookPost();
71         fb2.like = 1002;
72         fb2.comment = 14;
73         fb2.share = 290;
74         fb2.type = "Video";
75         fb2.genre = "Action";
76

```

```

77     FaceBookPost fb3 = new FaceBookPost();
78     fb3.like = 125;
79     fb3.comment = 10;
80     fb3.share = 2;
81     fb3.type = "Shorts";
82     fb3.genre = "Gameplay";
83
84     FaceBookPost[] fbPost = new FaceBookPost[3];
85     fbPost[0] = fb1;
86     fbPost[1] = fb2;
87     fbPost[2] = fb3;
88
89     Phone phone1 = new Phone();
90     phone1.brand = "iphone";
91     phone1.model = "iphone 16";
92     phone1.price = 250.99;
93     phone1.color = "red";
94     phone1.weight = 120;
95
96     Phone phone2 = new Phone();
97     phone2.brand = "Sumsung";
98     phone2.model = "Ultra s25";
99     phone2.price = 990.99;
100    phone2.color = "black";
101    phone2.weight = 100;
102
103    Phone phone3 = new Phone();
104    phone3.brand = "Vivo";
105    phone3.model = "Vivo Y17s";
106    phone3.price = 239.99;
107    phone3.color = "pink";
108    phone3.weight = 90;
109
110    Phone[] phone = new Phone[3];
111    phone[0] = phone1;
112    phone[1] = phone2;
113    phone[2] = phone3;
114

```

```

115    Book book1 = new Book();
116    book1.isbn = "Q11";
117    book1.title = "Samurai Japan";
118    book1.author = "Kim jungun";
119    book1.genre = "Politic";
120    book1.price = 20.99;
121
122    Book book2 = new Book();
123    book2.isbn = "QE231";
124    book2.title = "Kungfu panda";
125    book2.author = "Vireak rith";
126    book2.genre = "Comedy";
127    book2.price = 23.99;
128
129    Book book3 = new Book();
130    book3.isbn = "W3YU41";
131    book3.title = "Polpot";
132    book3.author = "Jame smith";
133    book3.genre = "History";
134    book3.prcie = 899.99;
135
136    Book book[] = new Book[3];
137    book[0] = book1;
138    book[1] = book2;
139    book[2] = book3;
140
141    }
142 }

```